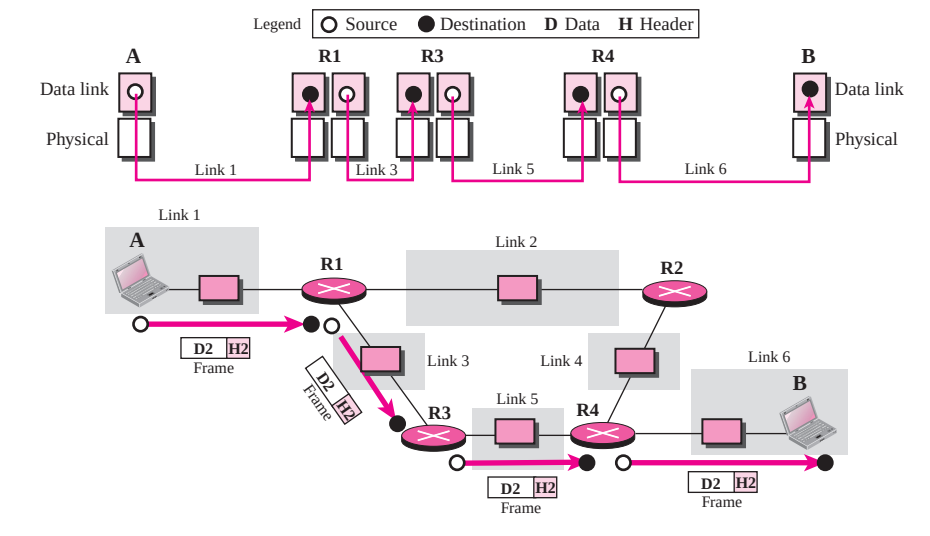


Figure 2.11 Communication at the data link layer

link layer protocol shown at the right to create a new frame to be sent to the router R3. The reason is that the two links, link 1 and link 3, may be using different protocols and require frames of different formats. Note also that the figure does not show the physical movement of frames; the physical movement happens only at the physical layer. The two nodes communicate logically at the data link layer, not physically. In other words, the data link layer at router R1 only *thinks* that a frame has been sent directly from the data link layer at computer A. What is sent from A to R1 is a stream of bits from one physical layer to another. Since a frame at A is transformed to a stream of bits, and the bits at R1 are transformed to a frame, it gives this impression to the two data link layer that a frame has been exchanged.

The unit of communication at the data link layer is a frame.

Network Layer

At the network layer (or, more accurately, the internetwork layer), TCP/IP supports the Internet Protocol (IP). The **Internet Protocol (IP)** is the transmission mechanism used by the TCP/IP protocols. IP transports data in packets called **datagrams**, each of which is transported separately. Datagrams can travel along different routes and can arrive out of sequence or be duplicated. IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination. Figure 2.12 shows the communication at the network layer.

Note that there is a main difference between the communication at the network layer and the communication at data link or physical layers. Communication at the